

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 8: Winds, Storms and Cyclones

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	8 — Winds, Storms and Cyclones
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: treating warm air as heavier, believing fast-moving air pushes outward (Bernoulli reversal), thinking wind blows from low to high pressure, swapping land/sea heating rates, calling the calm eye the most dangerous part, ranking a tornado as larger than a cyclone, and reading the can-crush as the metal folding itself rather than an outside-vs-inside pressure imbalance. Do not write on this paper — mark answers on your answer sheet.

1. A child claims that since a column of air several kilometres tall presses down on every rooftop, a flat roof of 10 square metres must carry a crushing load far greater than a parked car. Yet the roof does not collapse. The single most important reason it survives is that:

- (A) the weight of the air column is actually far too small to compare with a car
- (B) the roof material is special and immune to air pressure
- (C) air pressure only acts on liquids and living things, not on solid roofs
- (D) the same air also pushes upward and sideways on the structure, and the air inside the rooms pushes outward, so the downward push is very nearly balanced by opposing pushes

2. A sealed flask of air is connected by a thin tube to a beaker of coloured water; the tube dips below the water surface. A student gently warms the flask with her hands and bubbles rise out of the tube into the water; she then lets the flask cool and water creeps up into the tube. From this she should conclude that, for the SAME trapped air:

- (A) warming created new air that escaped as bubbles, and cooling destroyed air so water filled the gap
- (B) the water itself expanded into bubbles when warm and shrank when cool
- (C) warming made the air expand and push out (bubbles), while cooling made it contract so the outside pressure pushed water up the tube
- (D) the glass flask grew larger when warm, releasing air, then shrank to suck water in

3. Two table-tennis balls hang on threads a small gap apart. A student blows steadily through the gap and they swing together; a second student then sets a small electric fan blowing across only the OUTER face of one ball. Predict that ball's motion and the reason:

- (A) it swings outward, away from the other ball, because the fast air now on its outer side lowers the pressure there while the still air in the gap pushes it out
- (B) it still swings inward, because moving air always pulls objects toward whatever it is near
- (C) it does not move, because air streams cannot affect a hanging ball from the side
- (D) it swings outward because the fan physically blows the ball away like a push of wind

4. Three towns lie in a line: P at 1004 units of pressure, Q at 1000 units, and R at 988 units, with equal spacing between neighbours. Using only the high-to-low rule and the idea that a bigger pressure drop over the same distance gives faster wind, the strongest wind between adjacent towns blows:

- (A) from P toward Q, because P has the highest pressure of all three
- (B) from R toward Q, because air always moves toward the middle town
- (C) from R toward P, since wind moves from the lowest to the highest pressure
- (D) from Q toward R, because the 12-unit drop between them is the largest over equal spacing

5. In a can-crush demonstration, Group A boils water to fill the can with steam before sealing, while Group B simply caps a can of ordinary room-temperature air; both cans are then plunged into ice water. Group A's can crumples dramatically and Group B's barely changes. The best explanation is that:

- (A) the steam in A condenses to a tiny drop of liquid, leaving almost no gas, so its inside pressure falls far more than B's, where the cooled air still nearly fills the can
- (B) boiling made A's metal soft, so it folds under any cooling while B's stays rigid
- (C) B's can contained extra air that resisted the crush, while A's can had a vacuum from the start
- (D) ice water cools steam but cannot cool ordinary air, so only A's can responds

6. The Sun heats equatorial air far more than polar air, while it also heats land faster than the neighbouring sea. A student says the SAME basic rule explains both the giant equator-to-pole circulation and the small daily sea breeze. The shared rule is best stated as:

- (A) warmer air is heavier, so it sinks and pushes cooler air upward in both cases
- (B) wherever air is heated more strongly it expands, becomes less dense and rises, lowering the surface pressure there so cooler denser air flows in beneath it
- (C) sunlight directly pushes air sideways from cold places toward warm places
- (D) the rotation of the Earth alone moves the air, with heating playing no part

7. A diver descends from the surface of a lake to a depth where the water above her presses much harder than at the top. By close analogy, the reason air pressure is LOWER on a mountain peak than in a deep valley is that:

- (A) the peak is closer to the Sun, which thins the air by heating it away
- (B) less air lies above the peak than above the valley, so a shorter, lighter column of air presses on the peak
- (C) air loses its weight entirely once you climb above the clouds
- (D) high places spin faster with the Earth, flinging the air pressure away

8. A paper bag is hung from each end of a light rod balanced on a pivot; a candle is then placed beneath one bag. After a while that bag rises and the rod tips. Replacing the candle with a block of ice under the same bag would instead make that bag:

- (A) rise even higher, because cold air is lighter than warm air
- (B) sink, because the air in it is cooled, contracts to a denser state and is heavier than the surrounding air
- (C) stay perfectly level, because temperature cannot change a bag's balance
- (D) rise, because ice gives off cold vapour that lifts the bag

9. A long-distance runner notices that when a fast lorry overtakes her on a narrow road, she feels tugged toward it for an instant. The same physics that drew the lorry-side air explains why two ships steaming side by side can collide. The common cause is:

- (A) the moving lorry and ships create a powerful suction by removing all the air around them
- (B) moving air and water exert higher pressure in the gap, squeezing the bodies apart and then back
- (C) the fast-moving air (or water) in the narrow gap is at lower pressure, so the higher outside pressure pushes the bodies together
- (D) the lorry and ships carry a magnetic field that attracts nearby metal and water

10. A coastal village expects a sea breeze each afternoon and a land breeze each night. During a long total solar eclipse near midday, the sky darkens for several minutes and the land suddenly stops being heated. During that darkness the afternoon sea breeze would be expected to:

- (A) strengthen, because darkness always increases wind
- (B) reverse instantly into a strong land breeze, because the land becomes colder than the sea within seconds

(C) weaken, because the land stops out-heating the sea, so the pressure difference that drives the sea breeze fades

(D) stay exactly the same, because sea breezes do not depend on heating

11. A geography student notes that the summer monsoon brings rain from sea to land, while the winter winds blow dry from land to sea. He reasons that if, in some imaginary year, the land somehow stayed COOLER than the sea throughout the summer, the summer rains would:

(A) fail or weaken, because the sea-to-land wind depends on the land being warmer and lower in pressure than the sea

(B) be even heavier, because cool land always attracts more moist sea wind

(C) be unchanged, because the monsoon direction is fixed and never depends on temperature

(D) reverse to bring rain from the land out to sea instead

12. On a sweltering humid afternoon, towering dark clouds build and a thunderstorm breaks; on a cold, dry day no such storm forms. A student lists three needed ingredients for the storm: warm air that can rise fast, plenty of water vapour, and room overhead for tall clouds to grow. Which single change would most reliably PREVENT the storm?

(A) replacing the warm, moist surface air with cold, dry air that cannot rise quickly or supply much vapour

(B) increasing the amount of water vapour in the warm rising air

(C) making the surface air even warmer so it rises faster

(D) providing more clear height above for clouds to build into

13. During one thunderstorm a girl counts about 9 seconds between each lightning flash and its thunder; an hour later she counts only about 3 seconds. Both flashes and their thunder are made at the same instant. The shrinking gap tells her that:

(A) the storm has moved farther away, since nearer storms have longer gaps

(B) the storm has moved closer, so the sound now has a shorter distance to travel before reaching her

(C) the lightning has grown weaker, which is what shortens the delay

(D) the speed of sound has increased during the storm, closing the gap

14. A student sketches a sea-breeze diagram with arrows but forgets the labels. He knows: it is daytime, the air over the land is rising, and wind blows from high to low pressure. Which combination of facts must be true on his diagram?

(A) the land air is higher in pressure than the sea air, and the surface arrow points from sea to land

(B) the land air is lower in pressure than the sea air, and the surface arrow points from land to sea

- (C) both regions have equal pressure, so the surface arrow could point either way
- (D) the land air is lower in pressure than the sea air, and the surface arrow points from sea to land

15. Wind blows fast and horizontally across the open top of a tall, narrow chimney while the room below is still. Smoke is drawn UP the chimney more strongly. If the wind suddenly drops to nothing, the smoke's upward draw would:

- (A) strengthen, because still air over the chimney sucks harder than moving air
- (B) stay the same, because wind over the top has no effect on smoke below
- (C) weaken, because without the fast wind the chimney top no longer has lowered pressure to draw the smoke up
- (D) reverse and push the smoke down into the room, because calm air presses downward

16. Two weather maps cover the same coast. On Map 1 the high-pressure and low-pressure centres are drawn close together; on Map 2 the same two pressure values sit far apart. Comparing the winds the two maps predict between those centres:

- (A) Map 2 predicts stronger wind, because the air has more room to speed up over the long distance
- (B) Map 1 predicts stronger wind, because the same pressure difference is squeezed over a shorter distance, giving a steeper gradient
- (C) both predict identical wind, because only the two pressure values matter, not their spacing
- (D) neither predicts any wind, because wind needs the centres to be touching

17. An inland desert and a nearby seaside town lie at the same latitude. Over a full day and night the desert swings between scorching afternoons and freezing nights, while the seaside town stays comparatively mild. The chapter's best explanation is that:

- (A) the desert receives far more sunlight than the coast at the same latitude
- (B) water heats and cools more slowly than land, so day and night breezes between the sea and the coast keep the seaside temperatures moderate
- (C) sea air has no weight, so it cannot store any of the day's heat
- (D) the sea has no influence at all, and the difference is pure chance

18. As a cyclone grows over a warm sea, rising moist air condenses, and the heat this releases warms the air further, deepens the central low and pulls in faster wind. If the cyclone drifts over a patch of unusually COOL sea water, the most likely effect on this self-feeding cycle is that it:

- (A) speeds up, because cool water condenses vapour faster and releases more heat
- (B) slows, because cooler water supplies less warm moist air, so less heat is released and the central low does not deepen as fast
- (C) is unaffected, because once a cyclone forms it no longer needs the sea
- (D) stops only because the cool water freezes the wind in place

19. A meteorology club ranks four whirling air systems by typical diameter: a tornado, a thunderstorm cell, a single cyclone, and the calm eye inside that cyclone. From SMALLEST to LARGEST across, the correct order is:

- (A) the whole cyclone, then a thunderstorm cell, then the eye, then the tornado
- (B) the eye, then the tornado, then the whole cyclone, then a thunderstorm cell
- (C) the thunderstorm cell, then the whole cyclone, then the tornado, then the eye
- (D) tornado, then the cyclone's eye, then a thunderstorm cell, then the whole cyclone

20. In a glass-tumbler experiment, a tumbler is filled to the brim, a stiff card placed over the mouth, and the whole thing inverted; the card and water stay put. A student then repeats it but only half-fills the tumbler before inverting, trapping some air above the water. The card is now MORE likely to fall because:

- (A) the smaller amount of water is heavier than a full tumbler of water
- (B) air pressure pushes only on full containers and ignores half-full ones
- (C) the trapped air pushes down on the water, adding to the water's weight, so the upward air pressure on the card may no longer be enough to hold it
- (D) the trapped air glues itself to the card and drags it down

21. A coastal district once lost thousands of lives to a cyclone but in a recent storm of similar strength lost very few. A reporter lists four changes since then. Which one is the chapter's MAIN reason for the drop in deaths?

- (A) the cyclones striking the coast have become far gentler over the years
- (B) the sea has stopped surging over the low coast during storms
- (C) almost everyone has permanently abandoned the coastal district
- (D) weather satellites and radar now give accurate warnings a day or two ahead, so people reach shelters in time

22. A student writes: 'Wind blows from cold regions to hot regions because cold air is attracted to heat.' A teacher says the conclusion about direction is often correct but the reason is wrong. The teacher's precise correction is that:

- (A) the student is completely right, because cold air is genuinely attracted to warmth
- (B) wind blows from high to low pressure; hot regions usually become low pressure as their air rises, so cooler higher-pressure air does flow toward them, but pressure, not attraction to heat, is the real driver
- (C) wind always blows from hot to cold, so both the direction and the reason are wrong
- (D) wind direction has nothing to do with temperature or with pressure

23. A balloon is sealed over a flask's mouth, the flask is dipped in hot water and the balloon swells; the same flask is then moved straight into ice water. A student predicts the balloon and is asked to also explain what happens to the PRESSURE of the air inside relative to the outside. The correct prediction is:

- (A) the balloon collapses inward, because the cooling air contracts and its pressure falls below the outside air pressure
- (B) the balloon swells further, because cold air expands to take up more room
- (C) the balloon stays as swollen as in the hot water, because cooling has no effect once it was warmed
- (D) the balloon collapses, but because the cold made the rubber stretch and pull inward on its own

24. An anemometer, a barometer and a thermometer all sit at one weather station. Over an afternoon the anemometer cups spin faster and faster while the barometer reading drops steadily. Taken together, these two trends most strongly suggest that:

- (A) clear, calm, settled weather is on its way, since wind and pressure are both behaving normally
- (B) a storm or low-pressure system is approaching, since falling pressure goes with strengthening wind
- (C) the air is simply getting hotter, which is what the spinning cups always indicate
- (D) rain is falling right now, which is the only thing a barometer can ever show

25. A student holds a strip of paper just below her lip and blows hard across its TOP; the strip lifts. She then turns the strip so she blows hard across its BOTTOM surface while the top stays still. The free end of the strip should now:

- (A) still lift up, because blowing on paper always raises it
- (B) be pushed downward, because the fast air is now below it at lower pressure while the still higher-pressure air above presses it down
- (C) stay perfectly still, because the side you blow on does not matter
- (D) lift up even more strongly, because blowing underneath gives extra lift

26. A tropical storm with a circular shape hundreds of kilometres wide, a calm low-pressure centre ringed by the fiercest winds, and great spiralling cloud bands is called a 'cyclone' near India. A friend in another country describes an identical storm but calls it a 'typhoon.' The best conclusion is that:

- (A) they are three different sizes of storm, with the typhoon always the largest
- (B) the typhoon is really a tornado, which is unrelated to a cyclone
- (C) they are completely unrelated weather events that merely look alike
- (D) they are the same kind of storm given different regional names in different parts of the world

27. A cyclone's central pressure is recorded falling lower and lower over a day while its strongest winds blow faster and faster. A second, separate cyclone over the same warm sea has a central pressure that is not nearly as low. Comparing the two, the second cyclone most likely has:

- (A) slower strongest winds, because its shallower central low pulls in the surrounding air less forcefully
- (B) faster strongest winds, because higher central pressure throws the air outward harder
- (C) exactly the same wind speed, because all cyclones spin at one fixed rate
- (D) no wind at all, because only the deepest low can move air

28. A ship sails directly through the centre of a large cyclone. Its log records: fierce wind from the east, then a stretch of eerie calm with patches of clear sky, then fierce wind again from the WEST. The clearest explanation of this pattern is that:

- (A) two different cyclones happened to pass over the ship one after the other
- (B) the cyclone stopped, rested, and then started up again from scratch
- (C) the leading eye wall gave the first fierce wind, the calm clear eye passed over, then the trailing eye wall gave fierce wind from the opposite side
- (D) the wind simply changed its mind for no physical reason as storms often do

29. A student argues that the calm, clear eye is the most DANGEROUS place in a cyclone because it sits at the very centre. A more careful student corrects this. The correct statement about danger across a cyclone is:

- (A) the eye is indeed the most violent part, since the centre always has the strongest winds
- (B) the calm eye and the eye wall are equally violent, since both lie near the centre
- (C) the outermost cloud bands are the most violent, while the eye wall is the calmest
- (D) the eye itself is calm; the ring of fastest winds and heaviest rain is the eye wall just around it, which is the most dangerous part

30. Over a single hot afternoon a patch of atmosphere above a sun-baked plain is warmed strongly from below for hours. Tracking the surface pressure under that patch over the afternoon, you would expect it to:

- (A) rise, because warmed air is heavier and presses down harder
- (B) fall, because the warmed air expands and rises away, leaving less air pressing down on the surface
- (C) stay unchanged, because heating the air cannot affect the surface pressure
- (D) drop to zero, because warming removes all the air from above the plain

31. Two identical sealed jars of air are prepared in a valley. One is carried to a high mountain peak; the other stays in the valley. On the peak the trapped air is the same amount as before. Comparing the NET strain on the two jars' walls:

- (A) the peak jar feels a net outward strain, because the outside air now pushing in is thinner while the inside air still pushes out as strongly as before
- (B) the peak jar feels a net inward squeeze, because mountains have stronger air pressure

- (C) both jars feel exactly the same strain, because sealed air never changes
- (D) the peak jar feels no strain at all, because air loses all its pressure on mountains

32. On the open sea a cyclone's fierce winds, very low central pressure and the heaping action of the storm together drive a great mound of sea water onto a low coast. A student asks why this storm surge floods so much worse than ordinary high tide. The best answer is that:

- (A) the surge is only slow rainwater that gradually collects on the low coast
- (B) the sea level drops during a surge, exposing dry land that later floods
- (C) the surge is fresh river water flowing backward from upstream
- (D) the cyclone's low pressure and powerful winds pile up a huge mass of sea water and drive it far inland, far beyond a normal tide

33. A cyclone that has been intensifying over the open ocean crosses the coastline and within a day weakens sharply. A student offers four reasons; choose the chapter's CORRECT one and reject the rest:

- (A) over land the storm becomes colder than outer space and freezes solid
- (B) the buildings on land act as a solid wall that physically blocks all the wind
- (C) over land the storm is cut off from the warm, moist sea air that supplied the heat and vapour feeding it
- (D) the eye instantly closes the moment the storm touches soil, stopping the wind

34. A flask of air is heated and an attached tube dips into coloured water; as the air warms, the water level in the tube is pushed DOWN, and as the air cools, the level rises back UP. A student claims this proves air 'is created by heat and destroyed by cold.' The accurate statement is:

- (A) air really is created by heat and destroyed by cold, exactly as the student says
- (B) the same air expands when heated, pushing the water down, and contracts when cooled, letting the water rise back
- (C) the water expands and contracts on its own while the air does nothing
- (D) the glass flask grows and shrinks, and the air plays no role at all

35. A student says: 'Cyclone, hurricane, typhoon, tornado, thunderstorm are just five names for the very same storm.' Identify the precise error in this statement:

- (A) cyclone, hurricane and typhoon are regional names for the same storm, but a tornado and a thunderstorm are different things, so they are not all the same
- (B) there is no error, since all five really are one and the same storm
- (C) the only error is that a thunderstorm should also be called a typhoon
- (D) the only error is that a cyclone is actually a kind of tornado

36. Inside a developing thunderstorm cloud, warm moist air rushes upward while water drops and ice fall back down through it. A student asks which of these is the chief ENGINE driving the storm's violence. The best answer is:

- (A) cold, dry air quietly sinking to the ground with no moisture involved
- (B) lightning forming first and then dragging the air upward behind it
- (C) the warm ground touching and heating the cloud directly from below
- (D) the rapidly rising warm, moist air together with the water and ice falling back through it

37. A weather forecaster sees the barometer at a port falling unusually fast through the morning and at the same time fishermen radio that the wind out at sea is rising. Putting these together, the forecaster's safest advisory is to:

- (A) announce that fine, settled weather is guaranteed for the rest of the day
- (B) tell the fishermen the changes are meaningless and to keep fishing far out
- (C) warn that a storm or cyclone may be approaching and advise boats to return, since falling pressure with rising wind signals a low-pressure system
- (D) predict that the temperature is about to drop below freezing at the port

38. A light ball hovers in the upward air stream from a vertical hair dryer and springs back when nudged sideways. If the dryer is tilted so the stream points at, say, 45 degrees, the ball tends to keep hovering WITHIN the slanted stream because:

- (A) the ball is weightless and simply floats wherever the dryer points
- (B) the air in the stream is sticky and glues the ball to the dryer's path
- (C) tilting the dryer adds a magnetic pull that holds the ball in line
- (D) wherever the fast stream is, its pressure is lower, so the slower higher-pressure air around it pushes the ball back into the stream along its new direction

39. When a cyclone warning is broadcast, a family debates four actions. Using cyclone-safety reasoning, which single action is BOTH safe and correctly explained?

- (A) store clean drinking water in advance, because the storm may contaminate or cut off the normal supply
- (B) throw open all doors and windows, so the strong wind can pass straight through the house
- (C) go to the seashore to watch and photograph the giant waves up close
- (D) shelter under the tallest lone tree in the open, because tall trees block wind best

40. A student summarises a cyclone's self-feeding cycle in four steps but jumbles them: (i) faster wind spirals in toward the centre; (ii) condensing vapour releases heat that deepens the central low; (iii) the Sun warms the sea so warm moist air rises; (iv) the deeper low and rising air keep more moist air coming up. Put them in the order in which they build on one another:

- (A) (iii), then (ii), then (i), then (iv)
- (B) (i), then (ii), then (iii), then (iv)
- (C) (ii), then (iii), then (iv), then (i)
- (D) (iv), then (i), then (ii), then (iii)

41. A teacher fills a tumbler to the brim, covers it with a card, inverts it, and the water stays. She then asks: if instead of plain card she used a small, very heavy flat metal plate of

the same size, what would most likely happen and why?

- (A) the plate would be held even more firmly, because heavier covers are gripped harder by air pressure
- (B) nothing changes, because the cover's weight never matters to this experiment
- (C) the plate floats away upward, because heavy metal is lifted by air pressure
- (D) the plate may fall, because its extra weight adds to the downward pull and may exceed the upward push of air pressure that held the light card

42. A region of warm, low-pressure air sits beside a region of cool, high-pressure air. A student is asked which way the surface air moves and what becomes of the warm air. The fully correct description is:

- (A) surface air flows from the cool high-pressure region toward the warm low-pressure region, while the warm air, being less dense, rises
- (B) surface air flows from the warm low-pressure region to the cool high-pressure region, while the warm air sinks
- (C) no surface air moves, and the warm air stays put because pressures cannot affect it
- (D) surface air flows toward the warm region, but the warm air also sinks because it is heavier

43. A coastal town with a daily sea breeze installs a tall row of mirrors that, on one test day, reflect away much of the sunlight before it can heat the land. Compared with a normal sunny day, that test day's afternoon sea breeze would most likely be:

- (A) stronger, because the mirrors add their own wind to the breeze
- (B) weaker, because with less sunlight heating the land the land-sea pressure difference that drives the breeze is reduced
- (C) unchanged, because sea breezes do not depend on how much the land is heated
- (D) reversed into a strong land breeze, because the mirrors make the land hotter than the sea

44. During the WINTER over much of India the prevailing wind blows from the land out toward the sea, and these winds bring little rain. A student is asked to explain BOTH the direction and the dryness together. The correct combined reason is:

- (A) in winter the land cools more than the sea, so higher pressure over the cool land pushes air seaward, and that land air carries little moisture, so it is dry
- (B) in winter the sea cools faster than the land, so wet air rushes from the sea onto dry land
- (C) winter winds blow straight up, so they bring snow rather than rain
- (D) winter winds are wetter than summer winds because cold air always carries more water

45. A girl times the gap between lightning and thunder during a storm and finds it staying steady at about 6 seconds for several flashes, then suddenly dropping toward 2 seconds. The steady gap followed by a sharp drop tells her that the storm:

- (A) stayed in one place the whole time, since the gap can only depend on the lightning's brightness
- (B) held its distance for a while and then moved noticeably closer, since a shorter gap means less distance for the sound to travel
- (C) moved steadily away, because a shrinking gap means the storm is retreating
- (D) had its thunder speed up, which is what shortened the gap rather than any movement

46. A student lists three ingredients a cyclone needs to form and grow over the ocean: warm sea water, plenty of water vapour rising and condensing, and a region of low pressure that wind can rush into. Which single missing ingredient would most directly STOP a forming cyclone from intensifying?

- (A) increasing the amount of water vapour available to rise and condense
- (B) deepening the low pressure at the centre even further
- (C) supplying more warm moist air to feed the rising column
- (D) removing the warm sea water, so little warm moist air rises and little condensation heat is released

47. A demonstration on uneven heating uses a single lamp shining at a slant on a globe so the equator gets nearly head-on light and the poles get only grazing light. A student concludes that this slant explains the great wind belts of the Earth because it makes:

- (A) the poles warmer than the equator, so the wind belts blow from equator toward the poles only
- (B) every part of the globe equally warm, so no wind belts can form at all
- (C) the equator far warmer than the poles, so warm equatorial air rises and cold polar air sinks, setting up giant circulating winds that carry heat
- (D) the equator lose its air's weight, which is what starts the wind belts

48. A student watches a cyclone forecast and asks why a deep, very low-pressure centre matters so much for the strength of the wind. The clearest cause-and-effect answer is:

- (A) a deep central low makes the surrounding air still, so the wind dies down
- (B) a deep central low pushes the surrounding air outward, which is what we feel as wind
- (C) the deeper the central low, the bigger the pressure difference pulling in surrounding air, so the faster the wind that rushes toward the centre
- (D) the centre's pressure has no link to the wind speed, which is set only by the sea temperature

49. A student says that because air is invisible and 'empty,' it cannot really hold heat or have weight, so a balloon weighs the same whether it is empty or full of air. Using the chapter's ideas, the correct response is that:

- (A) air has no weight, so the student is right that a full and empty balloon weigh the same
- (B) air has weight only when it is moving as wind, not when it is still in a balloon
- (C) a full balloon weighs less than an empty one, because air makes things lighter
- (D) air does have weight, so a balloon full of air weighs more than an empty one, and that weight is also why air exerts pressure

50. A weather buff records four facts about one storm: a roughly circular shape only a few hundred metres across, an extremely fast whirling column reaching down from a cloud to the ground, a very short life of minutes, and great local destruction along a narrow path. This storm is best identified as a:

- (A) full tropical cyclone hundreds of kilometres wide
- (B) calm sea breeze blowing in from the coast
- (C) tornado
- (D) wide, slow-moving high-pressure region

51. A teacher heats a flask of air sealed by a balloon, and the balloon swells; she asks the class to predict what would happen to the SURROUNDING room air's role if the flask were instead cooled below room temperature. The correct prediction is:

- (A) the inside air would expand on cooling, so the balloon would swell even more
- (B) the room air would do nothing, because only the flask's own air can affect the balloon
- (C) the inside air would freeze solid and the balloon would simply pop off
- (D) the inside air would contract and lower its pressure, so the higher-pressure room air would push the balloon inward

52. A roof builder in a cyclone-prone area is told that fast wind across a roof can tear it off. He proposes three designs. Using the fast-air-low-pressure idea, which design BEST resists being lifted, and why?

- (A) a very light, loosely laid roof, because light roofs are too small for the wind to lift
- (B) a tall, broad roof left unanchored, because its size alone holds it down
- (C) a heavy, well-anchored, low-profile roof, because firmly fixing it down counters the upward lift created when fast wind lowers the pressure above it
- (D) any roof at all, because fast wind over a roof actually presses it down, not up

53. A student lights a candle just inside the lower opening of a tall glass chimney standing on a table; the flame burns brightly with a steady upward draught. If a card is slipped over the top opening to block it, the flame would most likely:

- (A) burn brighter, because the trapped warm air feeds the flame more
- (B) be unaffected, because a flame does not depend on air moving through the chimney
- (C) turn into smoke that rises faster, because blocking the top speeds the draught
- (D) dim or go out, because blocking the top stops the warm air rising out and fresh air drawing in, cutting off the draught the flame needs

54. A student claims a cyclone's eye must be the windiest place because it is exactly at the centre where 'everything is strongest.' To test this, she reads that the eye is calm and clear while the surrounding eye wall is fiercest. The correct general lesson is that:

- (A) the centre of any storm is always automatically the most violent point
- (B) the calm eye and the violent eye wall must really be the same thing under two names
- (C) being at the geometric centre does not make a place the most violent; in a cyclone the violence peaks in the eye wall ring around the calm eye
- (D) the outermost edge of a cyclone is always windier than any inner ring

55. A teacher pours hot water into a metal can to fill it with steam, seals it, and cools it; the can crumples. She then asks what would happen if, just before cooling, she had punched a large hole in the can and left it open while it cooled. The can would:

- (A) not crumple, because outside air would flow in through the hole to equalise the pressure, so no inside-outside difference could build up
- (B) crumple even harder, because the open hole lets the steam escape faster
- (C) crumple only on the side with the hole, while the rest stays straight
- (D) melt around the hole instead of crumpling anywhere

56. A coastal town gets a steady afternoon sea breeze on most clear days. On one heavily overcast afternoon the clouds keep the land from heating much, and on another day a cold spell leaves the land actually colder than the sea. On BOTH such days you should expect the afternoon sea breeze to:

- (A) weaken or even reverse, because the land-warmer-than-sea condition that drives the sea breeze is reduced or flipped
- (B) strengthen on both days, because cloud and cold both increase wind
- (C) stay exactly the same on both days, since the sea breeze never depends on land temperature
- (D) become a stronger sea breeze on the cold day, because cold air rushes toward the warm sea as a sea breeze

57. A student maps India's two wind seasons: a moist sea-to-land summer monsoon and a dry land-to-sea winter wind. She is asked what single fact most fundamentally underlies BOTH reversals across the year. The deepest correct answer is:

- (A) land changes temperature faster than the sea, so in summer the hotter land draws sea air in and in winter the cooler land pushes air out to sea
- (B) the Moon's gravity reverses the winds between summer and winter
- (C) the sea changes temperature faster than the land, which sets both wind directions
- (D) the monsoon winds simply reverse on a fixed calendar with no link to heating

58. A student watches a thunderstorm and writes down four events in a scrambled list: (i) drops grow heavy and fall as rain; (ii) warm moist air rises and cools; (iii) water vapour

condenses into towering tall clouds; (iv) the ground and surface air are warmed on a hot humid day. Put them in the order they actually occur:

- (A) (i), then (ii), then (iii), then (iv) (B) (iv), then (ii), then (iii), then (i)
(C) (iii), then (i), then (iv), then (ii) (D) (ii), then (iv), then (i), then (iii)

59. Two cyclones, X and Y, are tracked over the same warm sea. X's central pressure keeps falling and its eye-wall winds keep speeding up; Y's central pressure has levelled off at a higher value. A forecaster must decide which is more dangerous to a coast and why. The sound conclusion is:

- (A) Y is more dangerous, because its higher central pressure throws out stronger winds
(B) they are equally dangerous, because all cyclones over the same sea behave identically
(C) X is more dangerous, because its deeper and still-deepening low pulls in faster eye-wall winds than Y's shallower, steady low
(D) neither is dangerous, because a cyclone's central pressure has nothing to do with its wind

60. A student is shown a cross-section of a cyclone from the calm centre outward and must label, in order, three zones by their wind and pressure. The correct centre-to-edge labelling is:

- (A) a violent, high-pressure eye, then a calm eye wall, then a still outer ring with no clouds
(B) a calm high-pressure eye, then calm clouds, then a violent outer rim that is the fiercest part
(C) a calm, low-pressure eye, then a violent eye wall of fastest winds and heaviest rain, then outer bands of spiralling clouds
(D) a calm eye, then a calm eye wall, then a calm edge, with no violent zone anywhere

Solutions — Science (Class 7), Chapter 8: Winds, Storms and Cyclones — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	A	21	D	31	A	41	D	51	D
2	C	12	A	22	B	32	D	42	A	52	C
3	A	13	B	23	A	33	C	43	B	53	D
4	D	14	D	24	B	34	B	44	A	54	C
5	A	15	C	25	B	35	A	45	B	55	A
6	B	16	B	26	D	36	D	46	D	56	A
7	B	17	B	27	A	37	C	47	C	57	A
8	B	18	B	28	C	38	D	48	C	58	B
9	C	19	D	29	D	39	A	49	D	59	C
10	C	20	C	30	B	40	A	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Air pressure acts equally in every direction, so the large downward force on the roof is matched by an upward force from below and by the indoor air pushing out; the pushes nearly cancel, which is why the roof is not crushed even though the air column genuinely is heavy. The idea that the column is too light to matter is wrong because the load really is large; it is the balancing, not the smallness, that saves the roof. No material is immune to air pressure, and air presses on solids just as it does on liquids and bodies.

2. (C) The trapped air is the same throughout; heating expands it so some is pushed out as bubbles, and cooling contracts it, lowering the inside pressure so the surrounding pressure drives water up the tube. No air is created or destroyed by temperature. The bubbles are escaping air, not boiling water, and the slight change in the rigid glass cannot account for the effect.

3. (A) Fast-moving air exerts lower pressure, so whichever side the fast stream is on becomes the low-pressure side and the still higher-pressure air on the opposite side pushes

the ball toward the stream; with the fan on the outer face, the ball is pushed outward. Moving air does not 'pull' by attraction, and a side stream certainly does affect a light hanging ball. The motion is caused by the pressure difference, not by the fan's air physically shoving the ball.

4. (D) Between adjacent equally spaced towns, the P-to-Q drop is 4 units while the Q-to-R drop is 12 units, so the steeper gradient Q-to-R drives the fastest wind, blowing from higher (Q) to lower (R). Merely having the highest pressure does not make P's gradient the steepest. Wind always moves from high to low pressure, so it cannot blow from R toward Q or from R toward P.

5. (A) Steam condensing to liquid removes nearly all the gas, producing a far larger inside-pressure drop than merely cooling air, so A crushes hard while B, whose air only contracts a little, barely dents. The metal is not softened by boiling; both cans are equally rigid. B does not start with extra air, nor does A start with a vacuum, and ice water cools air just as readily as steam.

6. (B) Both the planetary circulation and the local breeze arise because more-heated air expands, becomes less dense, rises, and leaves a low-pressure region that denser cooler air flows into. Warm air is lighter, not heavier, so it rises rather than sinks; sunlight warms air rather than shoving it sideways; and although the Earth turns, the question's scope attributes both flows to uneven heating, not to rotation alone.

7. (B) Just as deeper water carries a taller column of water pressing down, a low valley has more air above it than a peak does, so the valley feels a heavier air column and higher pressure while the peak feels less. Nearness to the Sun does not 'thin air away,' air does not lose its weight at altitude, and the effect is about the height of the air column, not the spin of the Earth.

8. (B) Cooling makes the air in the bag contract and become denser, so it is heavier than the surrounding air and the bag sinks, the opposite of the candle case where warmed, less-dense air made the bag rise. Cold air is denser and heavier, not lighter; temperature clearly does change the balance as the candle case proves; and ice does not emit a lifting vapour.

9. (C) Where air or water moves fast, its pressure drops, so the still higher-pressure fluid on the outer sides pushes the two bodies toward each other across the low-pressure gap. There is no total removal of air, the moving fluid's pressure is lower (not higher) in the gap so it cannot push them apart, and no magnetic attraction is involved.

10. (C) The sea breeze is driven by the land heating faster than the sea, which lowers the land's surface pressure; if the heating briefly stops, that pressure difference fades and the breeze weakens. Darkness does not by itself increase wind, the land's stored warmth means it cannot become colder than the sea within mere minutes to drive an immediate reversal, and the breeze certainly does depend on differential heating.

11. (A) The moisture-laden sea-to-land summer wind blows because the strongly heated land develops lower pressure than the sea; without that heating, the driving pressure difference is gone and the rains fail or weaken. Cool land does not attract more moist wind, the monsoon direction is not fixed but follows the heating, and land carries little moisture so a reversed land wind would be dry, not rainy.

12. (A) A thunderstorm needs warm air that rises rapidly carrying abundant vapour to condense into tall clouds; removing the warmth and moisture removes the engine and the storm cannot form. Adding more vapour, making the air warmer, or giving more room overhead all encourage the storm rather than preventing it.

13. (B) Light reaches the eye almost instantly while sound lags, so the time gap measures how far the sound travelled; a shorter gap means the storm is nearer. Closer storms give shorter gaps, not longer; the brightness or strength of the lightning does not set the delay; and the speed of sound does not change to close the gap, the distance does.

14. (D) Because the land air is rising it has left lower pressure at the land surface, and since wind blows from high to low pressure the surface wind must blow from the higher-pressure sea toward the lower-pressure land. Rising air cannot leave higher pressure behind, a sea-to-land breeze cannot come from a land-to-sea arrow, and equal pressures would produce no definite breeze at all.

15. (C) Fast wind across the chimney top lowers the pressure there, so the higher pressure below pushes the smoke up; remove the wind and that low-pressure draw disappears, so the upward pull weakens. Still air does not suck harder than moving air, the wind plainly does affect the smoke, and calm air does not actively push smoke back down the chimney.

16. (B) Wind speed depends on how steeply pressure changes with distance; the same pressure difference packed into a shorter gap gives a steeper gradient and faster wind, so Map 1's winds are stronger. More room does not let wind 'speed up' more; spacing matters as much as the values; and centres need not touch for wind to blow.

17. (B) The sea's slow heating and cooling, combined with the daily exchange of sea and land breezes, moderates the coast's temperatures while the inland desert, far from any such buffer, swings widely. Same-latitude places receive comparable sunlight, sea air does have weight and does store heat, and the sea's moderating influence is real, not coincidental.

18. (B) The cycle is fed by warm, moist air from the sea; over cooler water less warm vapour is supplied, less condensation heat is released, the low deepens more slowly and the storm weakens. Cool water gives the storm less, not more, energy; a cyclone keeps needing the warm sea to sustain itself; and the wind is not 'frozen,' it simply loses its energy source.

19. (D) A tornado is a narrow violent column, the eye is a small calm core perhaps tens of kilometres across, a thunderstorm cell is larger, and the whole cyclone spans hundreds of kilometres, so smallest to largest runs tornado, eye, thunderstorm cell, cyclone. The other orders wrongly make the cyclone smallest, place the tornado as larger than the eye is meant to be reversed, or scramble the relative sizes.

20. (C) With trapped air above, that air presses down on the water and adds to the downward load, so the upward air pressure on the card may be overcome and the card falls; in the full case the upward air pressure alone exceeds the water's weight. Less water is lighter, not heavier; air pressure acts regardless of how full the tumbler is; and the air does not adhere to and pull the card down.

21. (D) Modern satellite and radar warnings give enough advance notice for people to evacuate to shelters, which is the chief reason cyclone deaths have fallen even when storm strength is similar. The storms have not become gentler, storm surge still floods low coasts, and people still live along the coast, so those cannot be the main reason.

22. (B) Pressure is the true driver: a hot region's rising air leaves low pressure, so denser cooler high-pressure air flows in, which often coincides with the cool-to-hot direction the student noticed, but heat does not 'attract' air. The student's reason is wrong even when the direction happens to match; wind does not always blow hot-to-cold; and direction is fundamentally tied to pressure differences set up by temperature.

23. (A) Cooling contracts the trapped air, dropping its pressure below the outside pressure, so the higher outside pressure pushes the balloon down into the flask. Cold air contracts rather than expands, cooling certainly does reverse the swelling, and it is the air's lowered pressure, not the rubber acting on its own, that drives the balloon inward.

24. (B) A steadily falling barometer signals an approaching low-pressure system, and faster-spinning anemometer cups confirm strengthening wind, the two together pointing to an oncoming storm. Falling pressure with rising wind is the opposite of settled weather; the anemometer measures wind speed, not temperature; and the barometer measures pressure, not current rainfall.

25. (B) Whichever surface the fast air sweeps across becomes the low-pressure side; with the fast air below, the higher-pressure still air above pushes the strip down. Blowing does not always lift paper, the side you blow across does matter, and air rushing beneath lowers the pressure there rather than adding lift.

26. (D) A cyclone, typhoon and hurricane are the same type of large rotating tropical storm under different regional names, so an identical description simply uses a different label. They are not fixed different sizes, a typhoon is not a tornado, and they are not unrelated, they are one phenomenon named variously.

27. (A) A deeper central low draws surrounding higher-pressure air in more forcefully, giving faster winds, so the cyclone with the less-deep low has slower strongest winds. Higher central pressure does not fling air outward, cyclones do not all spin at one rate, and a shallower low still moves plenty of air.

28. (C) Passing through the centre, a ship meets the front eye wall (fierce wind), then the calm clear eye, then the rear eye wall whose winds blow from the opposite side, exactly matching the log. It is one storm, not two; the cyclone does not pause and restart; and the reversal is a consequence of the eye's structure, not random.

29. (D) The eye is a calm, clear, low-pressure core, and it is the surrounding eye wall that holds the fastest winds and heaviest rain, making the eye wall the most dangerous part. The eye is not the most violent place, it is not as violent as the eye wall, and the outer bands are milder than the eye wall, not fiercer.

30. (B) As the air is warmed it expands, becomes less dense and rises away, so less air remains pressing on the surface and the pressure there falls. Warm air is lighter, not heavier; heating clearly does change surface pressure; and the air rises away rather than vanishing, so the pressure drops but does not reach zero.

31. (A) The sealed air inside pushes out as strongly as in the valley, but on the peak the thinner outside air pushes in less, so the jar feels a net outward strain. Mountain air pressure is lower, not stronger, so there is no extra inward squeeze; the surroundings do change even though the trapped air does not; and air does not lose all pressure at altitude.

32. (D) The combination of very low central pressure and fierce winds heaps the sea into a great mound that washes far inland, flooding the coast much worse than an ordinary tide. The surge is sea water driven by the storm, not slow rainwater; the sea level rises, not drops; and it is sea water, not backward-flowing river water.

33. (C) A cyclone feeds on warm, moist sea air; once over land that supply is cut off, so it loses energy and weakens within a day. It does not freeze, buildings cannot wall off a storm hundreds of kilometres wide, and the eye does not snap shut on contact with land, the loss of the moist-air supply is the cause.

34. (B) The same trapped air expands on heating, pushing the water down, and contracts on cooling, letting the water rise, with no air created or destroyed. The student's claim is wrong, the effect is driven by the air rather than the water acting alone, and the slight glass change cannot explain the clear level shifts.

35. (A) While cyclone, hurricane and typhoon are indeed regional names for one type of storm, a tornado is a small violent whirling column and a thunderstorm is yet another phenomenon, so lumping all five together is wrong. The statement is not error-free, a thunderstorm is not a typhoon, and a cyclone is not a kind of tornado.

36. (D) The violent up-and-down motion of rapidly rising warm moist air and falling water and ice is the engine of a thunderstorm's energy and its electric activity. Quiet sinking dry air does the opposite of driving a storm, lightning is a result of the churning rather than its cause, and the ground does not heat the high cloud by touch.

37. (C) A sharply falling barometer together with rising wind points to an incoming low-pressure storm, so the prudent step is to warn of a possible cyclone and call the boats back. Such readings are the opposite of settled weather, they are far from meaningless, and they indicate a pressure-driven storm rather than a freeze.

38. (D) The fast stream is always a low-pressure region, so the surrounding higher-pressure still air pushes the ball back toward the stream no matter which way it points, keeping the ball cradled in the slanted flow. The ball is not weightless, the air is not sticky, and there is no magnetic pull involved.

39. (A) Storing clean drinking water is wise because a cyclone can contaminate or interrupt normal supplies. Opening all doors and windows lets damaging wind and rain in rather than helping, going to the shore exposes people to the surge and waves, and a lone tall tree is dangerous in high wind and lightning rather than protective.

40. (A) The chain starts with the Sun warming the sea so moist air rises (iii), then condensation releases heat that deepens the low (ii), which makes faster wind spiral in (i), and that feeds yet more moist air upward (iv), looping the cycle. Beginning with the inflowing wind, the heat release, or the feedback step puts an effect before its cause.

41. (D) The upward air pressure on the cover holds up the water plus the cover's own weight; a heavy plate adds enough downward weight that the air pressure may no longer balance it, so the plate can fall. Air pressure does not grip heavier covers harder, the cover's weight does matter, and air pressure does not lift heavy metal upward and away.

42. (A) Wind blows from high to low pressure, so surface air moves from the cool high-pressure side to the warm low-pressure side, where the warm, less-dense air rises. Air does not flow from low to high pressure, the difference does drive a flow, and warm air rises rather than sinking because it is lighter, not heavier.

43. (B) The sea breeze grows from the land heating faster than the sea; reflecting sunlight away heats the land less, shrinking the pressure difference and weakening the breeze. Mirrors do not generate wind, the breeze does depend on land heating, and reflecting sunlight cools the land rather than making it hotter than the sea.

44. (A) In winter the land cools more than the sea, so the cooler land becomes higher pressure and pushes air out to sea, and because that air comes off the land rather than over the ocean it carries little moisture and stays dry. The sea does not out-cool the land in winter, the winds are horizontal rather than vertical, and land winds are drier, not wetter, than the moist summer monsoon.

45. (B) Because light arrives almost instantly and sound lags, the time gap measures distance; a steady gap means steady distance and a sudden drop means the storm moved closer. The gap depends on distance, not brightness; a shrinking gap means approach, not retreat; and the speed of sound does not change, the distance does.

46. (D) Warm sea water is the source of the warm moist air whose rising and condensation power the storm; remove it and the cyclone is starved of energy and cannot intensify. Adding more vapour, deepening the low, or supplying more warm moist air all strengthen the storm rather than stopping it.

47. (C) Head-on sunlight makes the equator much warmer than the grazing-lit poles, so warm equatorial air rises and cold polar air sinks, driving the planet-scale circulation that moves heat around. The poles are colder, not warmer; the heating is uneven rather than equal; and the equatorial air does not lose its weight.

48. (C) Air rushes from high pressure toward low pressure, so a deeper central low means a larger pressure difference and faster inrushing wind. A deep low energises rather than stills the wind, it draws air inward rather than pushing it out, and the central pressure is directly linked to wind speed.

49. (D) Air genuinely has weight, so a balloon full of air is heavier than an empty one, and it is precisely this weight of the atmosphere that produces air pressure. Air is not weightless, its weight does not switch on only when it moves, and adding air makes a balloon heavier, not lighter.

50. (C) A narrow, very fast whirling column reaching from cloud to ground, only a few hundred metres wide, short-lived and cutting a narrow path of destruction, is a tornado. A cyclone is hundreds of kilometres across, a sea breeze is a gentle daytime coastal wind, and a high-pressure region is calm and broad, none matching the description.

51. (D) Cooling contracts the trapped air and lowers its pressure, so the higher-pressure surrounding room air pushes the balloon inward, the reverse of the heating case. Cooling contracts rather than expands the air, the surrounding higher-pressure air is exactly what pushes the balloon in, and the air does not freeze solid in this everyday demo.

52. (C) Fast wind over a roof lowers the pressure above it, creating an upward lift, so a heavy, well-anchored, low-profile roof best resists being torn off. A light loose roof is easiest to lift, a large unanchored roof offers more surface for lift, and the wind creates lift rather than pressing the roof down.

53. (D) The flame's draught works because warm air rises and leaves the top while fresh air is drawn in below; blocking the top halts that circulation, so the flame is starved of fresh air and dims or dies. Trapping warm air does not feed the flame more, the flame plainly depends on the through-flow of air, and blocking the top slows rather than speeds the draught.

54. (C) The eye sits at the centre yet is calm, while the eye wall around it is the most violent, showing that central position does not equal maximum violence. The centre is not automatically the most violent, the calm eye and the violent eye wall are distinct regions, and the outer edge is milder than the eye wall.

55. (A) With a large open hole, outside air rushes in to keep the inside and outside pressures equal, so no crushing pressure difference can develop and the can stays whole. An open hole prevents the crush rather than worsening it, the pressure difference cannot build on only one side, and cooling does not melt the metal.

56. (A) The sea breeze relies on the land being warmer and lower in pressure than the sea; cloud cover or a cold land removes or reverses that condition, so the breeze weakens or flips. Cloud and cold do not increase wind here, the breeze does depend on land temperature, and on the cold day air would tend to flow seaward as a land-type breeze, not a stronger sea breeze.

57. (A) Because land heats and cools faster than the sea, in summer the hot land becomes low pressure and draws in moist sea air, while in winter the cool land becomes high pressure and pushes dry air out to sea, so the same heating-rate difference underlies both seasons. The Moon does not drive the monsoon, it is the land (not the sea) that changes temperature faster, and the reversal follows heating rather than a fixed calendar alone.

58. (B) The hot day first warms the surface air (iv), the warm moist air rises and cools (ii), its vapour condenses into tall clouds (iii), and finally the drops grow heavy and fall as rain (i). The other orders put rain or cloud formation before the warming and rising that must come first.

59. (C) A deeper, still-deepening central low draws in faster winds, so X with the lower and falling pressure is the more dangerous storm. Higher central pressure means weaker, not stronger, winds; cyclones over one sea do not all behave identically; and central pressure is directly tied to wind strength.

60. (C) From the centre out, a cyclone has a calm low-pressure eye, then the violent eye wall with the fastest winds and heaviest rain, then outer spiralling cloud bands. The eye is low pressure and calm rather than violent and high pressure, the eye wall is the fiercest zone rather than the outer rim, and a cyclone certainly has a violent zone, namely the eye wall.